

SUVAS T V

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PROFESSIONAL SUMMARY

ECE undergraduate (CGPA: 9.69) with hands-on experience in RTL design, FPGA-based Edge AI, digital systems, and hardware acceleration. Proficient in Verilog HDL, SystemVerilog, VHDL, and Xilinx Vivado. Seeking internships in VLSI Digital Design, RTL Design, or FPGA Engineering.

EDUCATION

SRM Institute of Science and Technology, Tiruchirappalli
Bachelor of Technology in Electronics and Communication Engineering

2024 – 2028 (Expected)
CGPA: 9.69/10.00

Honey Bunch Senior Secondary School (CBSE), Pollachi
Class XII – 88% | Class X – 96.2%

TECHNICAL SKILLS

Programming Languages: C, Python

Hardware Description Languages: Verilog HDL, SystemVerilog

Hardware Design: FPGA Design, VLSI Design, Embedded Systems, Digital Logic Design, Edge AI

EDA & Development Tools: Xilinx Vivado, ModelSim, LTspice, Git

Hardware / Platforms: FPGA (Xilinx/AMD), Arduino

RELEVANT COURSEWORK

Digital Electronics • VLSI Design • Embedded Systems • FPGA Design

PROJECTS

Real-Time Cardiac Fatigue Detection using HRV on FPGA (Edge AI) *Verilog HDL, FPGA (Xilinx), Vivado*
[GitHub](#)

- **Engineered a Low-Power Hybrid Edge AI Architecture:** Designed a two-tier hardware system on a Xilinx Artix-7 FPGA (Nexys A7) for real-time cardiac fatigue and arrhythmia detection, utilizing an always-on Tier-1 HRV-MLP (~ 100 μ W) to dynamically trigger a Tier-2 1D CNN (~ 50 mW) only when anomalies are detected.
- **Model Design & Quantization:** Trained PyTorch/TensorFlow models for HRV feature classification and ECG morphological analysis, achieving up to 97.0% classification accuracy (100% AFib sensitivity) while quantizing weights to **Q8.8 fixed-point representation** for efficient RTL hardware execution.
- **RTL Hardware Synthesis & Resource Optimization:** Implemented synthesizable Verilog HDL code on Xilinx Vivado (83.33 MHz), achieving extremely low hardware utilization (**0.56% LUTs**, **0.06% FFs**, and **3.70% BRAM**) with optimal timing margin (+2.185 ns WNS).
- **Automated Verification & CI/CD Pipeline:** Integrated SystemVerilog Assertions (SVA) within Verilog testbenches to validate bit-exact parity between Python golden models and FPGA hardware outputs, automated via GitHub Actions pipelines.

Automated Street Lighting System using LDR and Arduino *Arduino, LDR Sensor, Embedded C* [GitHub](#)

- Designed an automatic lighting system that adapts to ambient light using LDR sensing and Arduino control logic; filed as an Indian patent application.
- Developed real-time switching firmware to reduce energy consumption in smart city environments.

Spiking Neural Network using LIF Neuron Model Simulation *Python, NumPy, MATLAB* [GitHub](#)

- Simulated a Leaky Integrate-and-Fire (LIF) spiking neural network to study neuromorphic computing principles.
- Analyzed temporal spike patterns and membrane dynamics for event-driven computation research.

ACHIEVEMENTS

Secured 2nd Prize in BUILD N BLIND, NIT Tiruchirappalli — a reverse-engineering challenge involving incomplete circuits and abstract functional descriptions, requiring creative circuit realization under communication constraints.

PATENT

Published Indian Patent Application — Automated Street Lighting System using Light Dependent Resistor (LDR) and Arduino, Application No. 202641024982 A

LANGUAGES

English • Tamil • Telugu • Hindi